MILITARY SPECIFICATION

COUNTERMEASURES SET AN/DLQ-5(V)

This specification is approved for use by Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the manufacture and acceptance of the AN/DLQ-5(V) Countermeasures Set, referred to herein as the set.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Methods of Preservation.
MIL-E-5400	Electronic Equipment, Aerospace, General Specification for.
MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-M-7793	Meter, Time Totalizing.
MIL-E-17555	Electronic and Electrical Equipment Accessories and Repair Parts; Packaging and Packing of.
MIL-T-18303	Test Procedures, Preproduction, Acceptance, and Life for Aircraft Electronic Equipment; Format for.

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-704	Aircraft Electric Power Characteristics.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Tests; Exponential Distribution.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-45662	Calibration System Requirements.
MS17321	Meter, Time Totalizing, Miniature, Digital 28 Volt DC.
MS27473	Connector, Plug, Electrical, Straight, Crimp Type, Bayonet Coupling, Series II.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

1356AS300	Countermeasures Set AN/DLQ-5(V).

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1356AS301 Countermeasures Set AN/DLQ-5(V) (Interface Control Drawing)

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Set description. The set is a single module that contains a noise source, traveling wave tube amplifier, power supply, and control circuitry.

3.1.1 Major components. The set shall consist of the following major components:

a. Traveling Wave Tube (TWT).

b. Power Supply.

c. Voltage Controlled Oscillator (VCO) Mode Generator.

d. 25 decible (dB) Output Sampling Directional Coupler.

3.1.2 Government furnished equipment. The set shall utilize the following associated equipment which shall be Government furnished (see 6.4):

a. Command Logic Card: Part Number 1356AS218

b. Component Assy AM Board: Part Number 1356AS202

c. Component Assy Noise Board: Part Number 1356AS183

d. VCO (4-8 GHz): Part Number 1356AS207-3

e. VCO (8-11 GHz): Part Number 1356AS208-3

3.2 First article. When specified, a sample shall be subjected to the first article inspection (see 4.4 and 6.3).

3.3 Interface.

3.3.1 External electrical and RF. The external electrical interfaces for the set shall be provided through connectors J1, J2, and J3, and located as shown on Drawing 1356AS300. Connector type and function shall be as shown in Table I.

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TABLE I. Electrical and RF interface types and functions.

Reference Designation	Connector Type	Function	Pin
J1 Input Power/Control/Monitor	MS[1/]	Cmd A Cmd B Cmd C Cmd D Cmd Standby +28 Vdc Gnd, 28 Vdc return AM Waveform SAM Sweep Swept Noise Sweep	A B C D E Z W K L J
J2	Type N Female	RF Output	
J3	SMA Female	RF Output Sample	

[1/] The J1 connector shall be a MS type and shall mate with MS27473E16B99S.

3.3.2 External mechanical. The external mechanical interface dimensions shall be as shown on Drawing 1356AS300. The mechanical interface shall enable direct installation into Government selected electronic platforms using existing mounting holes.

3.3.3 Internal interface with Government furnished equipment (GFE). The physical and functional interface requirements between the GFE and the contractor's design shall be as shown on Drawing 1356AS301.

3.4 Controls. The set shall incorporate the controls and functions listed in Table II, which are included on the Government furnished equipment, except for the RF TUNE, which shall be developed by the contractor. All controls shall be readily accessible.

3.5 Time meter. The set shall be provided with a time-totalizing meter in accordance with MIL-M-7793, Type MS17321-10.

3.6 Materials, processes, and parts. In the selection of materials, processes, and parts, fulfillment of major design objectives for the intended application specified herein shall be the prime consideration. In so doing, the following shall govern:

a. Microwave technology shall be considered and microwave items shall conform to requirements specified herein.

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b. Materials, processes,and parts requirements shall be in accordance with MIL-E-5400.

c. Contractor-designed (peculiar) microwave items shall meet the general intent of MIL-E-5400.

d. Approval for the use of nonstandard materials, processes, and parts shall be obtained in accordance with with MIL-E-5400.

3.7 Characteristics.

3.7.1 Performance. The set shall meet the performance requirements specified herein at the conditions of 3.8.

3.7.1.1 Input power. The set shall operate as required when supplied a power source in accordance with Table II and Figures 9 and 10 of MIL-STD-704, except as follows:

TABLE II. Controls and functions.

Control	Function
RF TUNE	Developed by the contractor. Adjusts the set's frequency
TWT SAT	Controls noise amplitude into TWT
SPT BW	Adjusts RF bandwidth of spot and instantaneous noise bandwidth for barrage and swept noise
BAR BW	Adjusts bandwidth of barrage noise
SWP BW	Adjusts swept noise bandwidth
SWP FRQ	Adjusts swept noise frequency
SWP MULT/LO/HI	Switches swept noise frequency between low and high range
BLNK MULT/LO/HI	Switches blink freuqency between low and high range
BLNK FRQ	Adjusts blink frequency
BLNK DTY	Adjusts duty cycle of blink waveform
SAM FRQ	Adjusts low end of SAM frequency range
SAM RATIO	Adjusts high end of SAM frequency range
SAM SWP	Adjusts sweep time from low frequency to high frequency
SAM DTY	Adjusts duty cycle of SAM waveform
SPT SAT	Adjusts spot noise frequency linearity

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a. Steady State Voltage 28.0 ±3 Volts direct current (Vdc).

b. Not greater than 15.0 amperes.

3.7.1.2 Under-voltage, fault protection, and reverse polarity.

3.2.1.2.1 Under-voltage. The set shall not be damaged by primary power voltages less than 25.0 Vdc, although specified performance is not required during low-voltage periods. The set shall automatically resume specified operation when the voltage returns within the specified limits.

3.7.1.2.2 Fault protection. The set shall contain electronic overload and thermal cut-out and time delay circuits to provide fault protection for both microwave tube and power supply. The protective functions shall automatically reset upon clearance of the fault.

3.7.1.2.3 Reverse polarity. Reverse polarity protection shall be provided.

3.7.1.3 Warm-up time. Within three minutes after application of primary power, the set shall achieve specified output power and all other performance requirements.

3.7.1.4 Frequency range. The operating frequency range shall be 5.4 to 10.5 Gigahertz (GHz).

3.7.1.5 Frequency stability. The frequency of the set shall be stable within ±10 MHz from 5.4 to 10.5 GHz, from -35°C to +54°C.

3.7.1.6 Termination mismatch. The set shall meet specified frequency range and RF output with an output voltage standing wave ratio (VSWR) of 1.5:1 nominal. The set shall not be damaged when operating with a VSWR of up to and including 2:1 over the specified frequency range and at any phase angle.

3.7.1.7 RF output power. Using a single TWT amplifier, the set shall provide a RF power output of not less than 20 watts between 5.4 and 10.5 GHz into a 50-ohm load with a VSWR of 1.5:1 ±10 percent varied through all phase angles.

3.5.1.7.1 RF output sample. When the output of the set is 20 watts, the power at the RF sample, J-3, shall be +18 dBm ±2.0 dBm.

3.7.1.8 Noise figure. The set's noise figure shall be not greater than 32 dB.

3.7.1.9 Electromagnetic interference. The set shall meet the CE01 and RE02 requirements of MIL-STD-461 for electromagnetic interference.

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3.7.1.10 Modes of operation. The set shall provide noise jamming of pulse or CW radars. The equipment shall transmit, with or without amplitude modulation, an internally generated radio frequency. The set shall include remote control functions to command the following modes:

a. Spot Noise

b. Blink Spot

c. Swept AM and Spot

d. Barrage Noise

e. Blink Barrage

f. Swept AM and Barrage

g. Swept Noise

h. Spare

i. Spare

j. Spare

k. Standby

3.7.1.10.1 Spot noise.

a. FM noise bandwidth: Adjustable from 2 to 20 MHz in the frequency band of 5.4 to 10.5 GHz.

b. FM noise deviation frequency components: 100 Hz to 1.0 MHz.

3.7.1.10.2 Barrage noise.

a. FM noise bandwidth: Adjustable from 2 to 20 MHz (with no sweep) in the frequency band of 5.4 to 10.5 GHz.

b. FM noise deviation frequency components: 100 Hz to 1.0 MHz (with no sweep).

c. Sweep waveform: Not less than 100 kHz sawtooth.

d. Sweep bandwidth: Adjustable from 10 to 500 MHz in the frequency band of 5.4 to 10.5 GHz.

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3.7.1.10.3 Swept noise.

a. FM noise bandwidth: Adjustable between 2 to 20 MHz (with no sweep) in the frequency band of 5.4 to 10.5 GHz.

b. FM noise deviation frequency components: 100 Hz to 1.0 MHz (with no sweep).

c. Sweep waveform: Adjustable from 1 Hz to 9 kHz sawtooth.

d. Sweep bandwidth: Adjustable from 10 to 500 MHz in the frequency band from 5.4 to 10.5 GHz.

3.7.1.10.4 Swept AM.

a.	Sweep range:	Adjustable from 10 Hz to 1.0 kHz.
b.	Minimum frequency:	Adjustable from 10 Hz to 1.0 kHz.
c.	Sweep ratio:	Adjustable from 2:1 to 10:1.
d.	Sweep period:	Adjustable from 0.4 to 15.0 secs
e.	Duty cycle:	Not less than 5 percent nor greater than 95 percent adjustable.
f.	Depth of RF modulation:	Not less than 40 dB.

3.7.1.10.5 Blink AM.

a.	Frequency range:	Adjustable from 0.1 to 100 Hz
b.	Duty cycle:	Not less than 5 percent nor greater than 95 percent adjustable.
c.	Depth of RF modulation:	Not less than 40 dB.

3.7.1.11 Spectral purity. All harmonic and spurious emissions both in-band and out-of-band, shall be suppressed more than 25 dB from the carrier level.

3.7.1.12 Video-frequency tolerance. All video frequencies, when set, shall be stable within ±10 percent over the -35°C to +54°C temperature range.

3.7.1.13 Programming. The set shall be programmed as follows:

a. The equipment shall be fully energized by the external application of the +28.0 ±3.0 Vdc power.

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b. Remote control of ECM modes during flight:

(1) An ECM mode shall be selected when the appropriate command signal (see Table III) of +28.0 ±3.0 Vdc is applied for not less than 50 ms.

(2) Output function conditions shall be as presented in Table IV.

TABLE III. ECM mode and command signal.

ECM Mode	Command Signal
Spot Noise	A
Blink Spot	B
Swept AM and Spot	AB
Barrage Noise	C
Blink Barrage	AC
Swept AM and Barrage	BC
Swept Noise	ABC
Spare	D
Spare	AD
Spare	BD
Standby	STANDBY/CLEAR

3.8 Environmental conditions.

3.8.1 Operational. The set shall meet all requirements of 3.7.1 prior to during, and after the following environmental conditions, separately or any combination thereof. The set shall not exhibit any temporary or permanent degradation as the result of exposure to any of the following environments:

a. Reduced pressure: 50,000 feet (equivalent pressure)

b. High temperature: +54°C

c. Low temperature: -35°C

d. Vibration: ±10.0 gravity units (g) 5-2000 Hz bandwith

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e. Shock: 25 g, 11 ms pulse

f. Explosion: Air/fuel atmosphere ratio 13:1, sea-level - 50,000 feet

TABLE IV. Output function conditions.

Output Function	Condition	
	Active	Inactive
Command A	+28.0 ±3.0 Vdc	0.0 ±0.2 Vdc
Command B	+28.0 ±3.0 Vdc	0.0 ±0.2 Vdc
Command C	+28.0 ±3.0 Vdc	0.0 ±0.2 Vdc
Command D	+28.0 ±3.0 Vdc	0.0 ±0.2 Vdc
Standby/Clear	+28.0 ±3.0 Vdc	0.0 ±0.2 Vdc
Swept waveform	0.0 ±0.2 Vdc to +4.5 ±0.5 Vdc	0.0 ±0.2 Vdc
AM waveform	0.0 ±0.2 Vdc to +4.5 ±0.5 Vdc	0.0 ±0.2 Vdc
Swept AM sweep waveform	0.0 ±0.2 Vdc to +4.5 ±0.5 Vdc	0.0 ±0.2 Vdc

3.8.2 Nonoperational. The set shall meet all requirements of 3.7.1 after the following environmental conditions, separately or any combination thereof. The set shall not exhibit any temporary or permanent degradation as the result of exposure to any of the following environments:

a. Thermal shock: -54°C to +71°C

b. Humidity: 95 percent relative humidity

3.9 Physical characteristics.

3.9.1 Dimensions. The length, width, and height of the set shall be in accordance with Drawing 1356AS300.

3.9.2 Weight. The weight of the set shall be not greater than 20 pounds.

3.10 Reliability. The reliability of the set shall be not less than 100 hours mean-time-between-failures (MTBF).

3.11 Maintainability. The mean-time-to-repair (MTTR) of the set shall be not greater than 0.5 hour when replacing major assemblies or shop replaceable units.

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3.12 Storage life. The storage life of the set shall be such that the set can be stored for five years and then meet the requirements specified herein.

3.13 Nameplate and marking. The nameplate shall be in accordance with Drawing 1356AS300. Interchangeable parts and assemblies shall be legibly marked. All markings shall be in accordance with MIL-STD-130.

3.14 Interchangeability. All assembly and subassembly level items of hardware that are subject to replacement and have the same part number shall be electrically, physically, and functionally interchangeable. The set shall meet the interchangeability requirements of MIL-STD-454, Requirement 7, and the electrical and mechanical interface requirements of 3.3.

3.15 Safety. The set shall meet the safety requirements of MIL-STD-454, Requirement 1.

3.16 Human performance/human engineering. Unless otherwise specified herein, all controls and adjustments shall be set during production and initial testing. Components and assemblies that have to be replaced to meet maintainability requirements as specified herein shall be clearly identified in the set. Removal and substitution of these items shall not be impeded by their position within the set assembly or by other components or structures. No special equipment, tools, or fixtures shall be required to effect a removal and replacement action.

3.17 Workmanship. Fabrication and assembly practices shall be in accordance with MIL-STD-454, Requirement 9.

3.17.1 Soldering practices. Soldering practices shall be in accordance with MIL-STD-454, Requirement 5.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection system. The contractor shall assure product conformance to the requirements, inspections, and tests specified herein. The contractor's quality assurance program shall be planned and used in a manner to achieve levels of quality and reliability commensurate with the intended application specified herein. The contractor's inspection system shall be subject to review and

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approval by the procuring activity (see 6.2.2). The contractor shall notify the procuring activity of planned testing not less than 10 days prior to testing.

4.1.2 Test plans. Unless otherwise specified in the contract or purchase order (see 6.2.1), detailed test plans in accordance with MIL-T-18303 shall be prepared by the contractor prior to the commencement of testing (see 6.2.2).

4.1.3 Test data. Unless otherwise specified in the contract or purchase order (see 6.2.1), the detailed test data from both the first article tests and acceptance tests shall be prepared by the contractor (see 6.2.2). Test data for each set offered for acceptance shall be in accordance with MIL-STD-831. This data shall identify all rejections from acceptance tests and corrective actions taken.

4.1.4 Failure records. Unless otherwise specified in the contract or purchase order (see 6.2.1), failure analysis and proposed corrective action shall be prepared by the contractor (see 6.2.2).

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article tests (see 4.5).

b. Quality conformance tests (see 4.6).

(1) Acceptance tests (see 4.6.1).

(2) Reliability tests (see 4.6.2).

4.3 Inspection conditions. Unless otherwise specified in the applicable test procedure, all tests and inspections shall be conducted under the following conditions:

a. Temperature (ambient): +23° ±10°C

b. Atmospheric Pressure: 725 +50/-115 mm of mercury (28.5 +2.0/-4.5 inches of mercury)

c. Humidity: 10 to 90 percent

4.3.1 External power. The external power shall be +28.0 ±3.0 Vdc applied to J1 and shall simulate airborne vehicle power transfer, when specified in the applicable test procedure.

4.3.2 RF output termination. Unless otherwise specified herein, the set's RF output shall be connected to a separate 50 ohm RF load termination when power is applied to the set.

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4.3.3 Test equipment accuracy. Unless otherwise specified herein, all test equipment used shall have an accuracy within 20 percent of the tolerance for the variable to be measured. In some cases where an open-ended minimum or maximum requirement is given, the test equipment used shall be accurate to within 5 percent of that requirement.

4.3.3.1 Maintenance and calibration. All test equipment shall be maintained and calibrated in accordance with MIL-STD-45662. Certification to this effect prior to all inspections and tests shall be prepared by the contractor (see 6.2.2).

4.4 First article sample. Unless otherwise specified in the contract or purchase order (see 6.2.1), the first article sample shall consist of two sets and shall be subjected to the first article tests specified herein. The first article sample shall be manufactured using the same materials, processes, and procedures proposed for production.

4.4.1 First article approval. Approval of the first article sample shall be by the procuring activity upon satisfactory completion of all tests. The procuring activity reserves the right to require an additional sample and test if any set fails to meet any requirement specified herein (see 6.2.1). Any production by the contractor before acceptance of the first article sample will be at the contractor's risk. Subsequent to first article approval, the contractor shall not change materials, processes, or procedures without prior approval of the procuring activity by an engineering change proposal (see 6.2.2).

4.5 First article tests. The first article sample shall be subjected to the inspections and tests of Table V and shall be performed in the sequence specified therein. The contractor shall prepare and include test reports (see 6.2.2).

4.5.1 First article failure analysis and corrective action. When specified in the contract or purchase order (see 6.2.1), the procuring activity shall be notified within one working day of a first article test failure and its probable cause. Procuring activity approval is required before implementation of any corrective action in first article products (6.2.2).

4.5.2 Disposition of first article test sample. When specified in the contract or purchase order (see 6.2.1), sets which have been subjected to the first article tests shall be reworked by the contractor by replacing all wear or damaged items. After reworking, the contractor shall resubmit the sets for acceptance.

4.6 Quality conformance tests.

4.6.1 Acceptance tests. Acceptance tests shall be conducted on each set offered for acceptance and shall consist of the tests specified in Table V.

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TABLE V. First article and quality conformance tests.

Examination or Test	Method Paragraph		
	Requirement Paragraph	First Article	Acceptance
1. Visual examinations			
a. Dimensions	3.9.1	4.7.1.1	4.7.1.1
b. Weight	3.9.2	4.7.1.1	4.7.1.1
c. Nameplate and markings	3.13	4.7.1.3	4.7.1.3
d. Interchangeability	3.14	4.7.1.4	4.7.1.4
e. Workmanship	3.17	4.7.1.5	4.7.1.5
2. Manufacturing burn-in test		4.6.3	4.6.3
3. Performance			
a. Warm-up and operational stability	3.7.1.3	4.7.2.1	4.7.2.1
b. Primary power voltage variation (25.0 Vdc)	3.7.1.1	4.7.2.6	
c. Primary power voltage variation (31.0 Vdc)	3.7.1.1	4.7.2.7	
d. Termination mismatch	3.7.1.6	4.7.2.2	
e. Undervoltage, fault protection, and reverse polarity	3.7.1.2	4.7.2.3	
f. Controls	3.4	4.7.2.4	4.7.2.4
g. Time meter	3.5	4.7.2.5	4.7.2.5
h. Electromagnetic interference	3.7.1.9	4.7.2.8	
i. Spot noise FM noise bandwidth	3.7.1.10.1a	4.7.3.1	4.7.3.1
j. Spot noise FM noise deviation frequency components	3.7.1.10.1b	4.7.3.2	4.7.3.2
k. Barrage noise FM noise bandwidth and sweep bandwidth	3.7.1.10.2a	4.7.3.3	4.7.3.3

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TABLE V. First article and quality conformance tests. - Continued

Examination or Test	Method Paragraph		
	Requirement Paragraph	First Article	Acceptance
l. Barrage noise FM noise deviation frequency components and sweep waveform	3.7.1.10.2b, 3.7.1.10.2c, 3.7.1.10.2d	4.7.3.4	4.7.3.4
m. Swept noise FM noise bandwidth and sweep bandwidth	3.7.1.10.3a, 3.7.1.10.3d	4.7.3.5	4.7.3.5
n. Swept noise FM noise deviation frequency components and sweep waveform	3.7.1.10.3b, 3.7.1.10.3c	4.7.3.6	4.7.3.6
o. Swept AM	3.7.1.10.4	4.7.3.7	4.7.3.7
p. Blink AM	3.7.1.10.5	4.7.3.8	4.7.3.8
q. Spectral purity	3.7.1.11	4.7.3.9	4.7.3.9
r. Video frequency tolerance	3.7.1.12	4.7.3.10	4.7.3.10
4. Environmental			
a. Operational	3.8.1	4.7.4.1	
b. Nonoperational	3.8.2	4.7.4.2	
5. Reliability 1/	3.10	4.6.2	4.6.2 1/

1/ Reliability tests shall be performed on first article sample units and, when specified in the contract or purchase order (see 6.2.1), other selected units which have passed all other acceptance testing (see 4.6.2).

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When specified by the contract or purchase order (see 6.2.1), sets shall be randomly selected from those which have passed acceptance testing and then be subjected to the reliability tests of 4.6.2.

4.6.1.1 Acceptance test data. The contractor shall prepare acceptance test data sheets (see 6.2.2) for each set offered for acceptance.

4.6.2 Reliability tests. Reliability tests shall be conducted on the first article sample and on randomly selected sets which have passed acceptance testing to ensure compliance with 3.10 (see 6.2.1). The reliability tests shall be in accordance with the Probability Ratio Sequential Test (PRST) Plan VIII C, Figure C-8, Appendix C of MIL-STD-781. If during reliability testing the total test hours and test failures plotted on the PRST Acceptance/Reject criteria for Test Plan VIIIC of MIL-STD-781 shows a reject situation, the procuring activity reserves the right to stop acceptance of sets anytime reject situation exists pending a review of the contractor's efforts to improve the set, the set's components and parts, and the set's workmanship (see 6.2.1).

4.6.2.1 Reliability failure criteria. In addition to the failure criteria of MIL-STD-781, a set shall be considered to have failed when any of the monitored characteristics perform outside the specified limits for that characteristic. Preventive maintenance measures or adjustments shall not be performed upon the sets during the period of the test.

4.6.2.2 Reliability data and test reports. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor shall prepare reliability test data and reports, including failure records, for each set subjected to reliability testing (see 6.2.2). The data and reports shall be in accordance with MIL-STD-781 and shall be prepared for review not later than 10 working days after completion of reliability testing.

4.6.3 Manufacturing burn-in test. Each set shall be operated under the conditions specified herein for a period of ten hours without failure. A failure shall be anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the set. This test shall be deleted if the reliability test includes a test on each set which requires at least ten hours of set on time. The test shall be run at ambient temperature and humidity conditions. Vibration shall be at any non-resonant frequency between 20 and 30 Hz at a level of ±3 gs. The set shall be vibrated (without vibration oscillators) for a period of ten minutes prior to the beginning of the ten hour period of operation. Where feasible, the set shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to the plane for five minutes. Where it is not feasible to vibrate the set in two directions, the vertical direction shall be used. During the ten hour period of operation following the ten minute vibration period,

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the set shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the ten minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future sets. A record shall be kept of all failures. The ten hour period specified above may be composed of two five hour periods to conform with standard working hours.

4.7 Inspection and test methods. Details of methods and procedures shall be in accordance with the approved test plan (see 4.1.1).

4.7.1 Visual examinations.

4.7.1.1 Physical characteristics. The set shall be inspected for conformance with 3.9.

4.7.1.2 Materials, processes, and parts. The set shall be inspected for conformance with 3.6.

4.7.1.3 Nameplate and marking. The set shall be inspected for conformance with 3.13.

4.7.1.4 Interchangeability. The set shall be examined for conformance with 3.14.

4.7.1.5 Workmanship. The set shall be examined for conformance with 3.17.

4.7.2 Performance tests. Unless otherwise specified in the contract (see 6.2.1), performance tests shall be performed to ensure that the set shall operate for conformance with 3.7.1. Input power shall be in accordance with 3.7.1.1 and the performance shall be continuously monitored during the conduct of tests.

4.7.2.1 Warm-up and operational stability test. The set shall be operated under the provisions and conditions specified in 4.3. The set shall be operated for a period of 30 minutes and the following shall be observed:

a. Input power (see 3.7.1.1).

b. Warm-up time (see 3.7.1.3).

c. Programming (see 3.7.1.13).

d. Frequency range (see 3.7.1.4).

e. Frequency stability (see 3.7.1.5).

f. RF output power (see 3.7.1.7).

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g. RF output sample (see 3.7.1.7.1).

h. Noise figure (see 3.7.1.8).

4.7.2.2 Termination mismatch. The set shall be connected to a mismatched load having a VSWR of 2:1 with phase angle between the resistive and the reactive load components varied between zero and 360 degrees. Input power shall be 28.0 ±3.0 Vdc and the RF TUNE shall be adjusted to 6.0 GHz and 9.5 GHz and the following observed to determine if the set meets the requirements of 3.7.1.6:

a. Input power (see 3.7.1.1).

b. Warm-up time (see 3.7.1.3).

c. Frequency range (see 3.7.1.4).

d. RF output power (see 3.7.1.7).

4.7.2.3 Under-voltage, fault protection, and reverse polarity. The set shall be tested for under-voltage and fault protection by application of 20 Vdc and nondamaging simulated faults. The operation of fault protecton functions shall meet the requirements of 3.7.1.2. Reverse polarity protection shall be demonstrated.

4.7.2.4 Controls test. The set's controls shall be tested under simulated operational conditions to ensure they meet the requirements of 3.4.

4.7.2.5 Time meter test. The set's time meter shall be tested under simulated operational conditions with running time observed and meet the requirements of 3.5.

4.7.2.6 Primary power voltage variation (+25.0 +0.1 -0.0 Vdc) test. The primary power voltage shall be adjusted to +25.0 +0.1 -0.0 Vdc and the following observed:

a. Input power (see in 3.7.1.1).

b. Warm-up time (see in 3.7.1.3).

c. Frequency range (see in 3.7.1.4).

d. RF output power (see in 3.7.1.7).

4.7.2.7 Primary power voltage variation (+31.0 +0.0 -0.1 Vdc) test. The primary power shall be adjusted to +31.0 +0.0 -0.1 Vdc and the following observed:

a. Input power (see 3.7.1.1).

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b. Warm-up time (see 3.7.1.3).

c. Frequency range (see in 3.7.1.4).

d. RF output power (see in 3.7.1.7).

4.7.2.8 Electromagnetic interference test. The power lead conducted emission test CE01 and the radiated emission electric field test RE02 of MIL-STD-462 shall be performed to ensure that the set meets the requirements of 3.7.1.9.

4.7.3 Modes of operation test. The following modes of operation shall be tested to verify conformance with 3.7.1.10.

4.7.3.1 Spot noise FM noise bandwidth test. The spot noise FM noise bandwidth shall be tested to verify conformance with 3.7.1.10.1a at two frequencies: 6.0 GHz and 9.5 GHz.

4.7.3.2 Spot noise FM noise deviation frequency component test. The spot noise FM noise deviation frequency components shall be tested to verify conformance with 3.7.1.10.1b.

4.7.3.3 Barrage noise FM noise bandwidth and sweep bandwidth test. The barrage noise FM noise bandwidth and sweep bandwidth shall be tested to verify conformance with 3.7.1.10.2a and 3.7.1.10.2d at two frequencies: 6.0 GHz and 9.5 GHz.

4.7.3.4 Barrage noise FM noise deviation frequency components and sweep waveform test. The barrage noise FM noise deviation frequency components and sweep waveform shall be tested to verify conformance with 3.7.1.10.2b and 3.7.1.10.2d.

4.7.3.5 Swept noise FM noise bandwidth and sweep bandwidth test. The swept noise FM noise bandwidth and sweep bandwidth shall be tested to verify conformance with 3.7.1.10.3a and 3.7.1.10.3d at two frequencies: 6.0 GHz and 9.5 GHz.

4.7.3.6 Swept noise FM noise deviation frequency components and sweep waveform test. The swept noise FM noise deviation frequency components and sweep waveform shall be tested to verify conformance with 3.7.1.10.3b and 3.7.1.10.3d.

4.7.3.7 Swept AM test. The swept AM shall be tested to verify conformance with 3.7.1.10.4.

4.7.3.8 Blink AM test. The blink AM shall be tested to verify conformance with 3.7.1.10.5.

4.7.3.9 Spectral purity test. The spectral purity of the set shall be tested to verify conformance with 3.7.1.11 in 0.5 GHz steps from 5.4 GHz to 10.5 GHz.

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4.7.3.10 Video frequency tolerance test. The video frequency tolerance of the set shall be tested to verify conformance with 3.7.1.12.

4.7.4 Environmental tests. The environmental tests of Table VI shall be performed in accordance with MIL-T-5422, except as modified herein. The right is reserved by the procuring activity to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements specified herein (see 6.2.1). The test specimens shall be allowed to stabilize at room ambient conditions prior to performance of measurements and inspections.

TABLE VI. Modification of test chamber conditions for temperature-altitude tests.

Table II of MIL-T-5422 step	Temperature (degree C)	Altitude (feet)	Time (hours)	Input Voltage	Operating/ Nonoperating
1	-35	ATM 1/	2	---	Nonoperating
2	-35	ATM	---	25.0	Operating
3	-35	50,000	---	31.0	Operating
4	-10	ATM	---	31.0	Nonoperating
5	+54	ATM	16	---	Operating
6	+54	ATM	4	31.0	---
7	Omit	---	---	---	---
8	Omit	---	---	---	---
9	0	50,000	1.25	31.0	Operating
9A	AMB 2/	AMB	2	---	Nonoperating
9C	0	50,000	1.25	31.0	Operating
10	+30	35,000	2	31.0	Operating
11	Omit	---	---	---	---
12	Omit	---	---	---	---
13	Omit	---	---	---	---

1/ ATM equals 1 atmosphere

2/ AMB equals ambient

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4.7.4.1 Operational environmental tests. The set's performance shall be tested and meet the requirements of 3.7.1 during and after exposure to the following environments to verify conformance to 3.8.

4.7.4.1.1 Temperature and altitude test. The set shall be tested in accordance with the temperature-altitude test of MIL-T-5422 Table II, as modified herein by Table VI.

4.7.4.1.2 Vibration test. The set shall be tested in accordance with the vibration test of MIL-T-5422, Curve I and Curve II.

4.7.4.1.3 Shock test. The set shall be tested in accordance with the shock test of MIL-T-5422, except as modified herein to no greater than 25 g.

4.7.4.1.4 Explosion test. The set shall be tested in accordance with the explosion test of MIL-T-5422.

4.7.4.2 Nonoperational environmental tests. The set performance shall be tested and meet the requirements of 3.8.2 after exposure to the following environments.

4.7.4.2.1 Thermal shock test. The set shall be tested in accordance with the thermal shock test of MIL-T-5422.

4.7.4.2.2 Humidity test. The set shall be tested in accordance with the humidity test of MIL-T-5422, except that two continuous cycles shall be used instead of 10.

4.8 Preshipment inspection. Prior to shipment, inspection shall ensure that preservation, packaging, packing, and marking for shipment are in accordance with the requirements of Section 5.

5. PACKAGING

5.1 Preservation-packaging, and packing. Unless otherwise specified in the contract or purchase order (see 6.2.1), the level of preservation-packaging, and packing shall be in accordance with MIL-E-17555 and MIL-STD-794, Level C. The method of preservation for level C shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794 and the method of packaging shall be in accordance with MIL-P-116.

5.2 Marking.

5.2.1 Special marking. Special marking shall be as specified in the contract or purchase order (see 6.2.1).

5.2.2 Normal marking. Unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

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5.3 Spare parts. Unless otherwise specified in the contract or purchase order (see 6.2.1), spare parts shall be preserved and packaged to the same specification as a complete set except that weight of spare parts will determine selection of style of the box container.

6. NOTES

6.1 Intended use. The AN/DLQ-5(V) Countermeasures Set is intended for use in electronic warfare test and evaluation programs and fleet training. The set will be utilized as a modular unit, easily combined with other target vehicles to provide electronic warfare configurations.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Number, title and date of this specification.

b. Responsibility for inspection (see 4.1).

c. Requirement for detail test plans (see 4.1.2).

d. Requirement for detail test data (see 4.1.3).

e. Requirement for failure records (see 4.1.4).

f. First article sample (see 4.4).

g. Additional tests and samples, if required (see 4.4.1).

h. First article failure analysis and corrective action (see 4.5.1).

i. Disposition of first article (see 4.5.2).

j. Sets to be selected for reliability test (see 4.6.2 and Table V).

k. Option to stop acceptance of set during reliability testing (see 4.6.2).

l. Requirement for reliability data and test reports (see 4.6.2.2).

m. Requirement for performance tests (see 4.7.2).

n. Environmental tests, if other than specified in 4.7.4.

o. Level of preservation-packaging and packing (see 5.1).

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p. Special marking (see 5.2.1).

q. Level of packing for spare parts if different than the set (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR-7-104.9(n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph No.	Data requirement title	Applicable DID No.	Option
4.1.1	Inspection System	DI-R-4803	
4.1.2	Test Plans and Procedures	UDI-T-21347	
4.1.3, 4.6.1.1	Test Reports	DI-T-2072	
4.1.4, 4.5.1.	Failure Analysis and Corrective Action Report	DI-R-5299C	
4.3.3.1	Maintenance and Calibration Procedures	DI-R-7065	
4.4.1	Engineering Change Proposal	DI-E-2037	
4.5	Test Report, First Article	UDI-T-21349	
4.6.2.2	Reliability Test Plan	DI-R-7033	
4.6.2.2	Reliability Test Reports	DI-R-7034	
4.6.2.2	Failure Summary and Report Analysis	DI-R-7041	

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

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6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of 2 sets. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test approval of the documents first article.

6.4 Government furnished equipment. The contracting officer should arrange to furnish the property listed in 3.1.2.

Preparing activity:
Navy - AS

(Project 5865-N034)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-85578(AS) Countermeasures Set AN/DLQ-5(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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